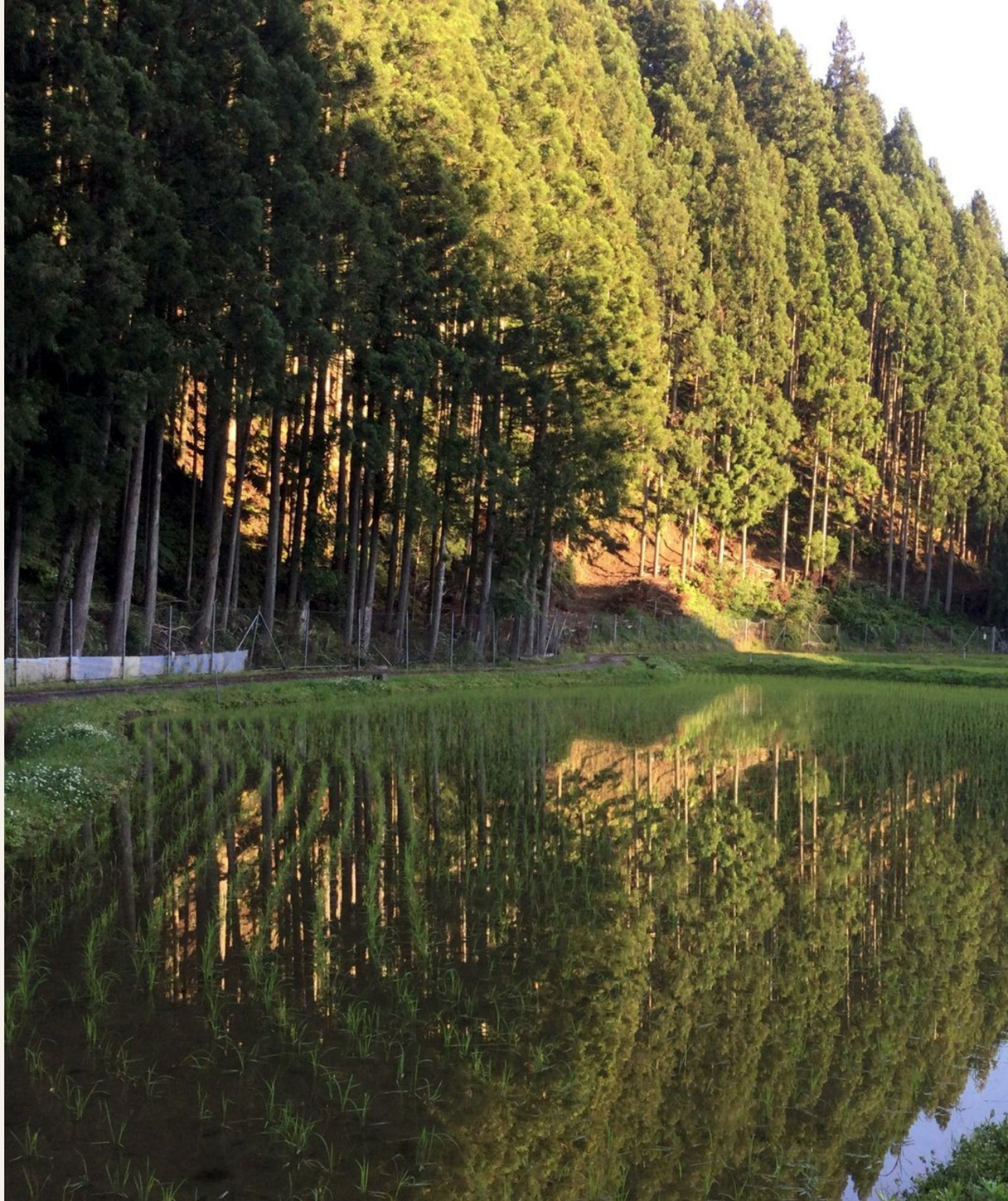
御 Mitsue Project
御杖プロジェクト

NAIST SEMINAR · 13 JANUARY 2027

Planted for a market that *no longer exists.*

いまは無い市場のために植えられた森

A forest, a village, and a sideways idea: history, today, and what we are still working out.



JAMES HILL · ROB OUDENDIJK

v0.3 DRAFT · 2026-11-09 · WORK IN PROGRESS

WHO IS TALKING

Two people, *one forest.*



James Hill

FOREST HISTORY · OPENS THE TALK

Takes us from the Tokugawa to 1958: how Japan came to plant this much sugi, and why it made sense at the time.

Rob Oudendijk

MITSUE-IT · THE REST OF THE TALK

Dutch engineer. Built his own house in the Mitsue forest, so the trees are 3 m from the front door. If the talk goes badly, blame the neighbours.

THE PLAN FOR 40 MINUTES

Forty minutes, *five stops.*

1 — *History*

How Japan came to plant so much sugi

JAMES · ~10 MIN

2 — *Today*

The village and forest as they are now

ROB · ~5 MIN

3 — *The problem*

Three problems with one shape

ROB · ~5 MIN

4 — *The idea*

Pull, don't push: wood, power, replanting

ROB · ~10 MIN

5 — *Homework*

What we checked and what we do not know

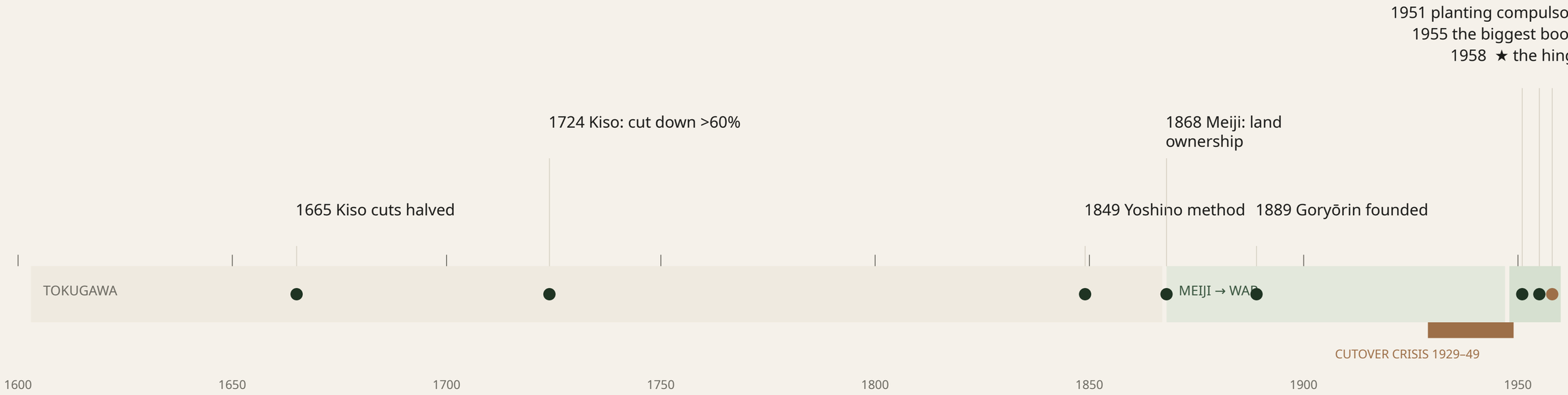
ROB · ~5 MIN + Q&A

Tone: light, not academic. Big scary problems sometimes have weirdly simple solutions, if you look at them sideways.

HISTORY 01 / 10 · 1603–1958 · THE WHOLE STORY IN ONE STRIP

JAMES · TO EXPAND

Three centuries, *one strip.*



The punchline arrives in 1958: the trees we see today were planted just as the demand behind them began to fade.

HISTORY 02 / 10 · 1603–1867 · TOKUGAWA TENURE

JAMES · TO EXPAND

Who owns a tree on shared land? *The question that made plantations possible.*

Land is feudal or communal. Private ownership is banned in principle; lords' forests in Kiso are guarded and theft is severely punished.

1665 and 1724. Kiso cuts halved, then cut down by more than 60% and held for 30 years.

Late 1600s: wave 1. Edo and Osaka boom; a 50-year timber investment pays for the first time.

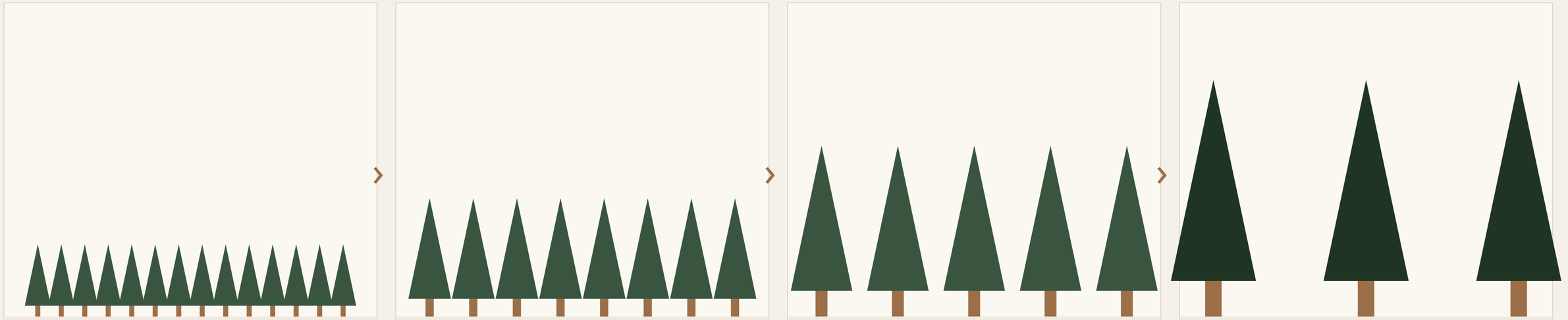
1801. Edo has about 1,000,000 people, bigger than London or Paris. With imports shut out, all demand falls on domestic forests.



HISTORY 03 / 10 · 1849 · THE YOSHINO METHOD

JAMES · TO EXPAND

Plant dense, thin early, *thin again.*



PLANT
at 3 ft spacing

THIN
when butt reaches 2–3 in.

RE-THIN
5 years later

FINAL
12 ft spacing, grow on

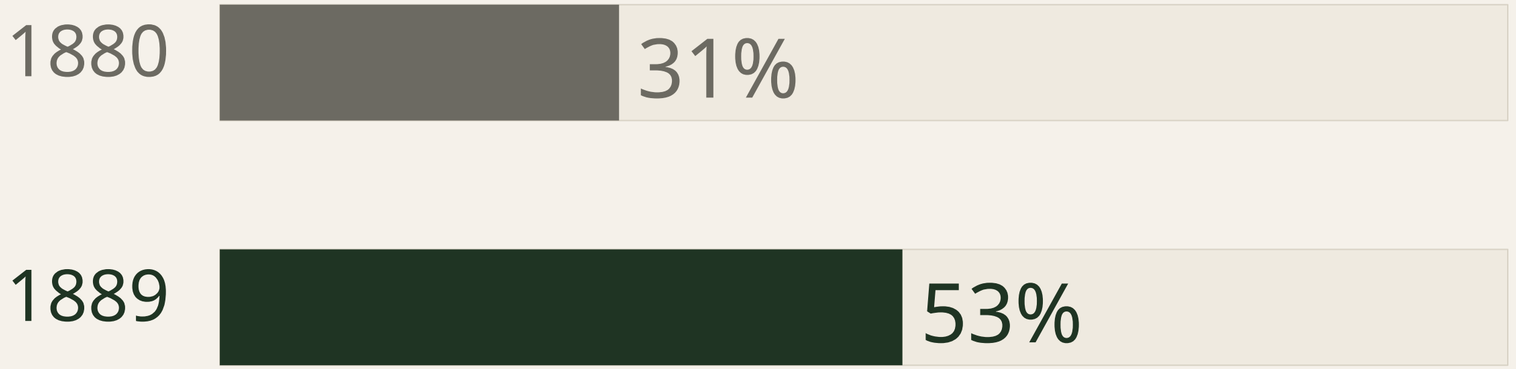
Okino Takao, Kurobane domain, 1849 *a citable Edo-era schedule that still looks familiar.*

Also: 1750s Yamaguni guild licence class for plantation growers; 1779 first Kiso forest inventory.

HISTORY 04 / 10 · 1868–1889 · KAN / MIN

JAMES · TO EXPAND

Government or private? *Six policy reversals in five years.*



share of Japan's woodland held by the government

1868 formal ownership introduced; national forests carved from communal and former feudal land.
1871–76 the state tries to sort all woodland into kan or min.

官有地 kan'yuchi

government. the state wants sanrin: dense conifer

民有地 min'yuchi

private. villagers need gen'ya: scrub for fuel and fodder

共有地 koyuchi

shared use. a third category, because the first two did not fit real life

HISTORY 05 / 10 · 1889–1947 · GORYŌRIN

JAMES · TO EXPAND

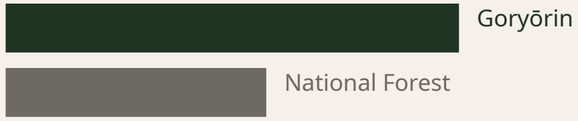
For 58 years, forestry was a *real business.*

1.6 M ha
carved out in 1889–90 as the Imperial Household's own income-producing forest estate

40 %
net margin on gross income at the 1936–45 peak, against 23% for the much larger National Forest

31 %
of Imperial Household income in 1940 (19% in 1916)

70 %
of Household wealth by 1946: ¥2.6 bn of cumulative asset value



1889 founded

1892 turns profitable

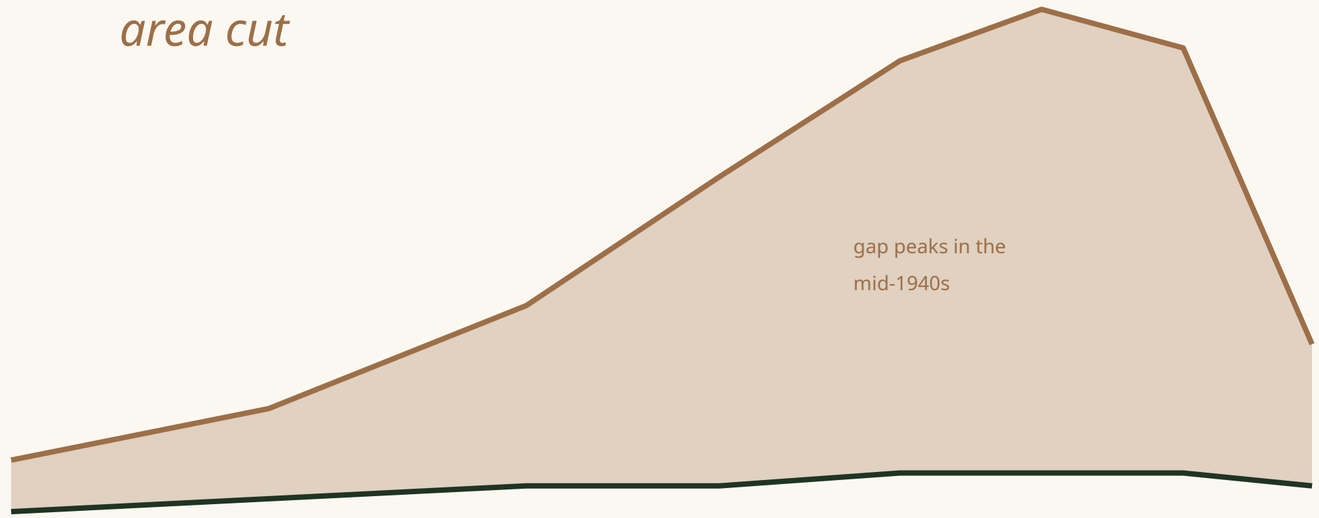
1942 Hundred Year Plan suspended, 'special harvesting'

1947 dissolved

HISTORY 06 / 10 · 1929–1949 · THE CUTOVER CRISIS

JAMES · TO EXPAND

Twenty years in a row, more was cut *than planted.*



1.49 M ha

cutover land in 1948, about 6% of Japan's forest area, concentrated near villages

8.6 → 10.97

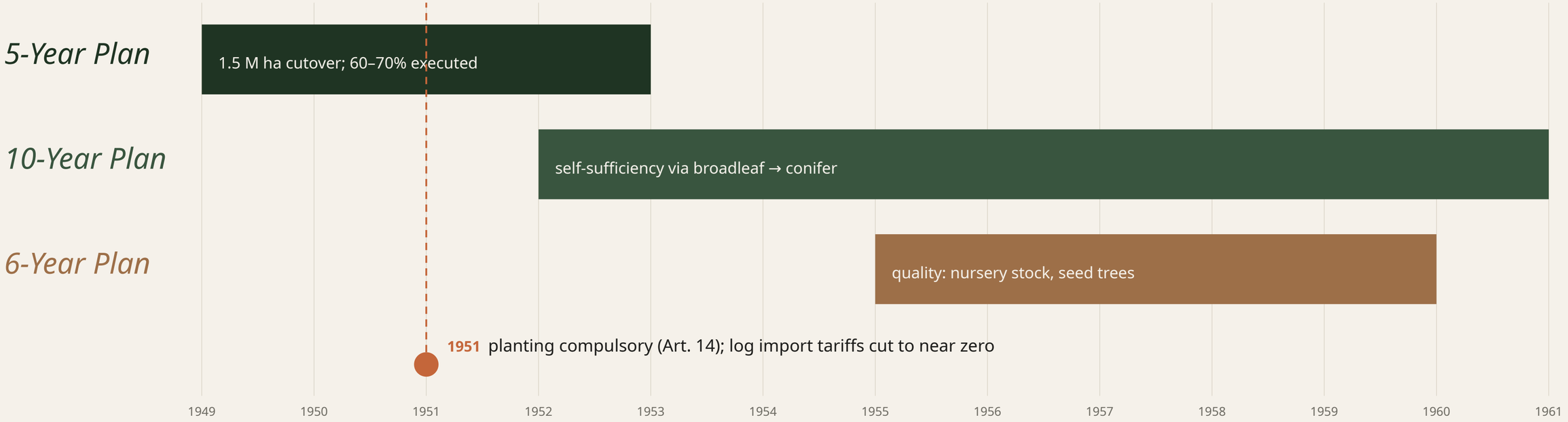
million koku harvested, 1942 to 1944: paid for by cutting reforestation and aftercare

1948 forecast: wood exhausted in 15–33 years. Wrong, but it drove everything after.

HISTORY 07 / 10 · 1949–1961 · POSTWAR PLANS

JAMES · TO EXPAND

Three plans in twelve years, *and planting becomes the law.*

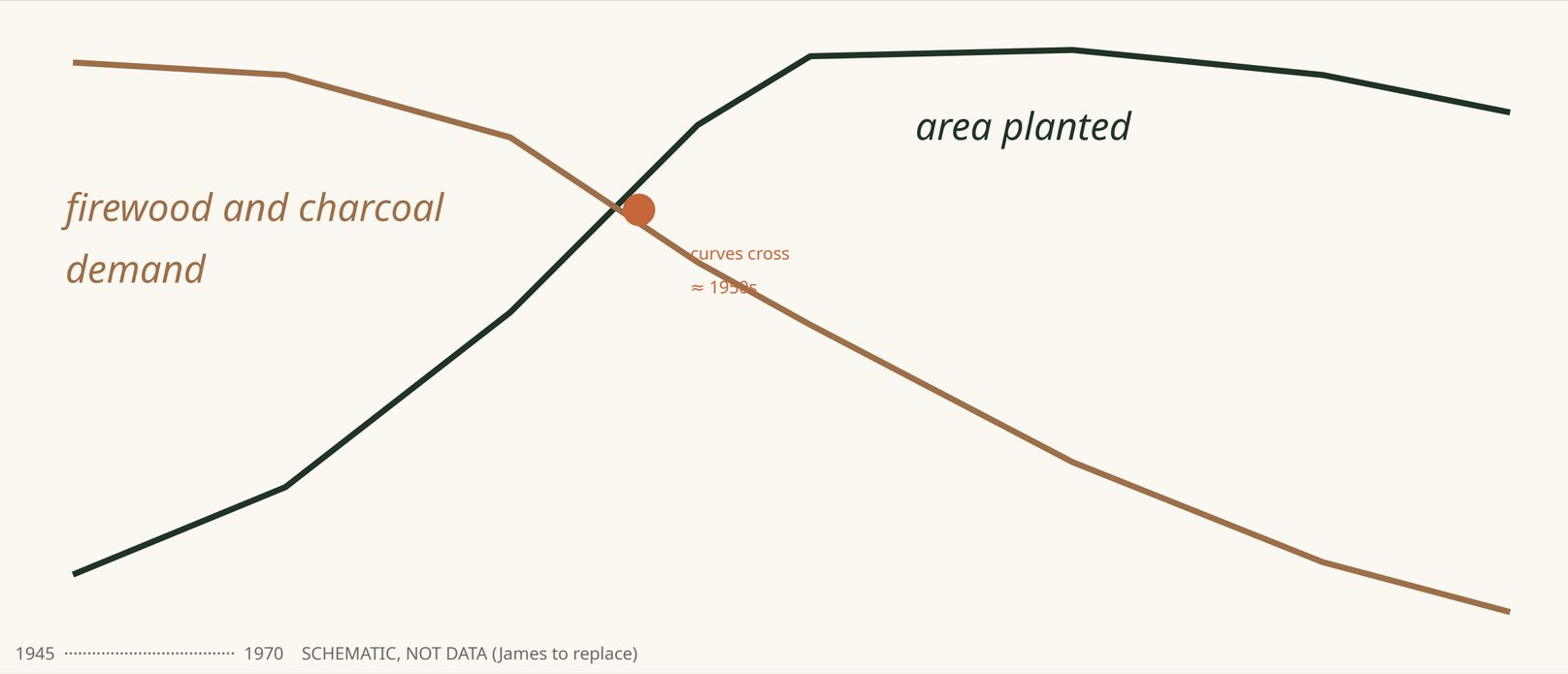


A tidy irony: broadleaf-to-conifer conversion is what we now want to undo, where it makes sense.

HISTORY 08 / 10 · 1950–1955 · TWO CURVES CROSS

JAMES · TO EXPAND

Planting peaks. *The reason for planting fades.*



1950

The demand floor drops out

Fossil fuels replace firewood and charcoal; chemical fertiliser replaces green fertiliser; machines replace draft animals.

1954–55

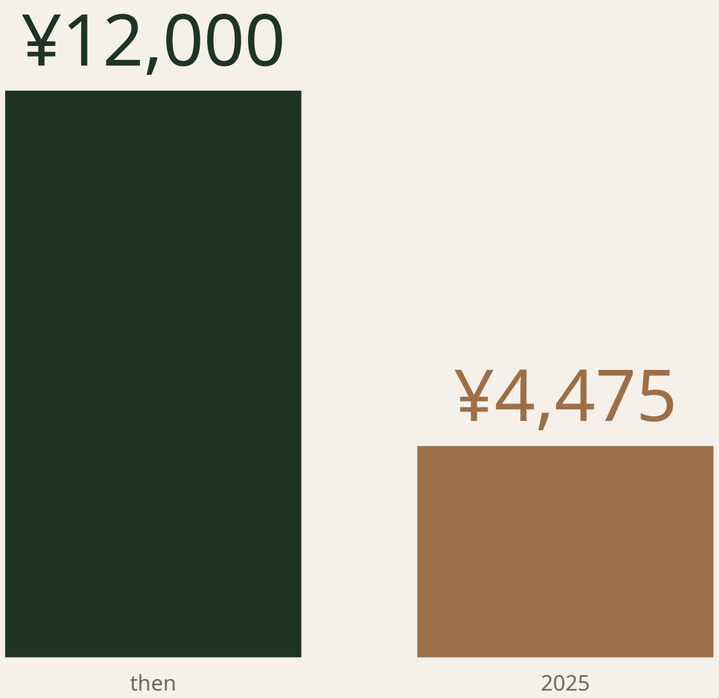
Peak planting

Highest annual area under the 10-Year Plan in 1954. Wave 3, the biggest boom, runs 1955–1970.

HISTORY 09 / 10 · 1951–2025 · SEEDS AND THE CRASH

JAMES · TO EXPAND

Planted as a retirement fund, *then the market crashed.*



A-grade sugi per m³ (years and series to be sourced)

1951

Import tariffs to near zero

Sensible at the time; it set up self-sufficiency to fall for decades.

1956

Forest Development Corporation

Long-term debt plants the least attractive land against a harvest 40–50 years out. After 1980 all 47 prefectural corporations fall into debt distress.

1970s–80s

Cheap imports

The market crashes before the planted trees are old enough to cash in. Only A-grade sells today.

JAMES · TO EXPAND

1958

The trees are older than *the demand that justified planting them.*

Deep inside the biggest boom (1955–70), planting compulsory since 1951, self-sufficiency still about 95%. The sugi and hinoki in Mitsue's mountains were planted in exactly this window. Over to Rob: today.



ROB, NOVEMBER 1958



TODAY · THE SETTING

A village of 1,400 people, *a forest of 7,000 hectares.*

≈ 1,400

residents

≈ 93 %

forest cover

≈ 7,050 ha

forest, over 90% private

50–60 yr

age of the main sugi and hinoki stands

Many owners are elderly or live elsewhere. At today's prices a cut sugi barely pays for the cutting. Still checking: how many owners, how many parcels, who inherits.

THE PROBLEM

Three problems with one shape: *nobody is paid to fix it.*



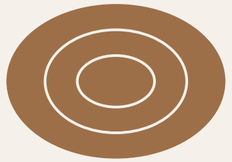
Not enough hands

The cooperative is small and has few young people. Harvest at scale needs machines and trained operators. Thinning is planned, but most of it is cut and left.



A ticking clock

A large share of the forest reaches harvest age inside the same ~25-year window in which new small-scale power technology matures. Nice timing, if we use it.



No pull to replant

Cutting sugi for cheap lumber does not pay enough to put anything back. Short-term profit, repeat forever. Nobody has a reason to break the cycle.

THE PROBLEM IN NUMBERS

STILL RESEARCHING

Most of the thinning wood *stays in the forest.*

18,140 m³ cut and left in the forest

2,574 m³ taken out

PLANNED THINNING FY2024–26 · 113 PARCELS · ≈ 222 ha · 4 DISTRICTS



≈ 88 %

stays in the forest: about 860 m³ a year actually comes out

That cut-and-left wood is a fuel resource, if it can be collected at a price that makes sense. Steep slopes and missing roads are why it is left.

Still checking: how much is reachable with a winch-assisted machine, and what an earlier '120–300 m³/yr' cooperative figure really measured.

Sources: Village forestry plan R6–R8 (FY2024–26); village hall (Oba)

THE IDEA

Pull, don't push: *give the wood a buyer that pays.*

1 *Thin and clear*

Cut-to-length machines on steep sugi

2 *Sort A / B / C / D*

A and B to the sawmill, C and D to chips

3 *Chip and dry*

Local chipping, drying with CHP heat

4 *Small CHP*

Gasification plant: power and heat, 24 h









Why 'pull'. Waiting for a subsidy pushes wood out once, then stops. A buyer that wants fuel 24 hours a day pulls wood out every year, and can pay for the replanting.

Then replant. Cleared stands go back to native broadleaf (oak, chestnut, konara) for wildlife and watershed, with fast-growing kiri on part of the land. Species chosen with the NGO more trees.

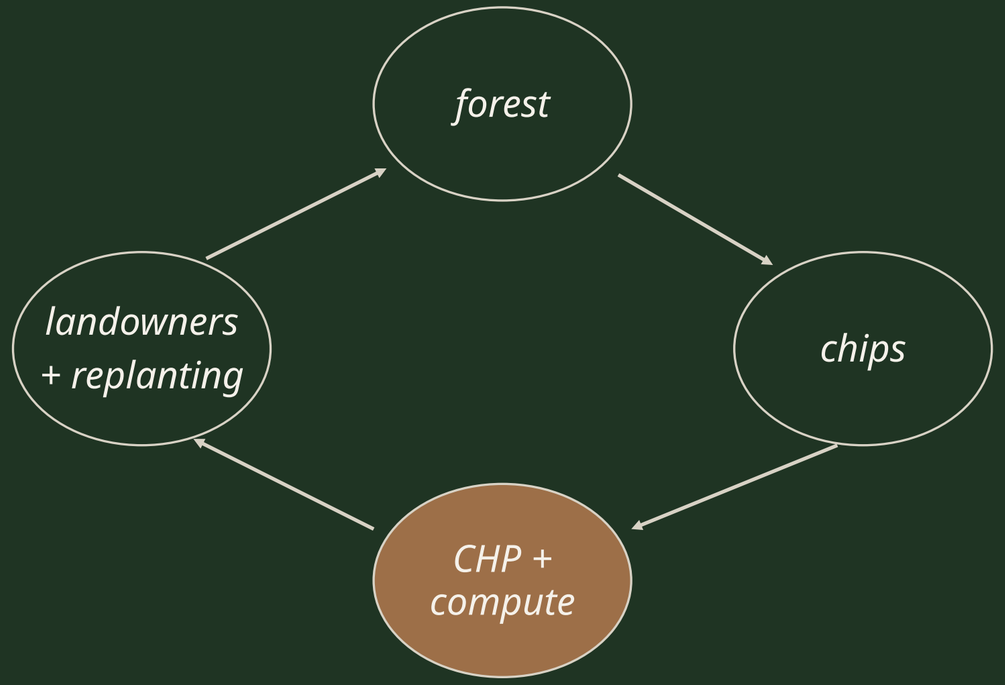
THE IDEA · TWO WAYS TO MOVE WOOD

Push stops. *Pull keeps going.*

PUSH · A ONE-OFF SUBSIDY



PULL · A BUYER FOR FUEL, 24 HOURS A DAY



Money from the plant flows back to owners and pays for replanting: the loop closes.

THE IDEA · WHAT A TREE IS WORTH

STILL RESEARCHING

Only A-grade wood sells. *B, C and D are a problem we can use.*

A



Straight, clean, few knots

Sawlog. The only grade worth cutting for lumber.

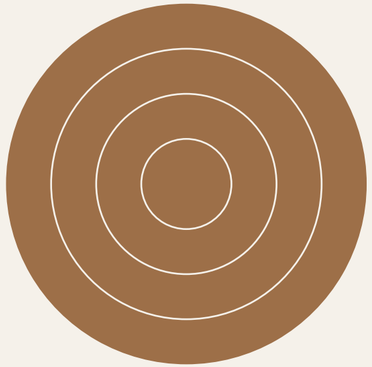
B



Slightly bent, some knots

Sawlog or laminated. Price depends on the buyer.

C



Crooked or small

Chips, pulp, board. Cheap.

D



Branches, tops, bark

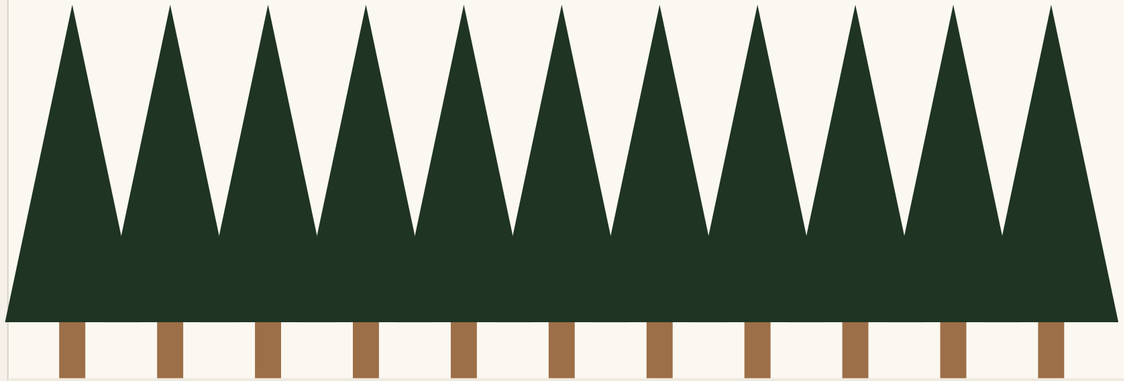
Today: left on the slope.

Sources: Grade descriptions simplified; model v1.2 (61% A national average, 22% A thinning-type stands)

THE IDEA · THEN REPLANT

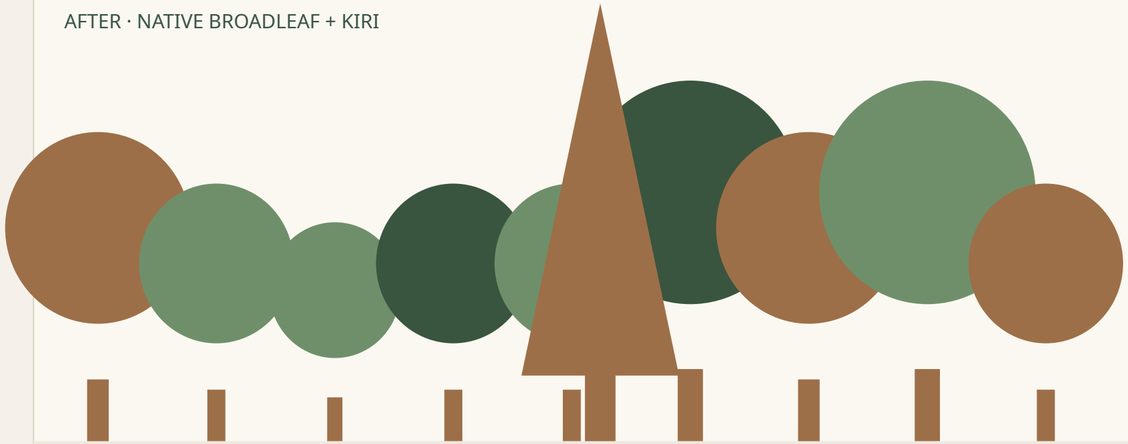
From one crop of cedar *to a forest that feeds the bears.*

BEFORE · MONOCULTURE



pollen, landslide risk, nothing to eat

AFTER · NATIVE BROADLEAF + KIRI



oak, chestnut, konara: acorns for wildlife

Species are chosen with the NGO more trees. The village's default is native broadleaf, not another monoculture.

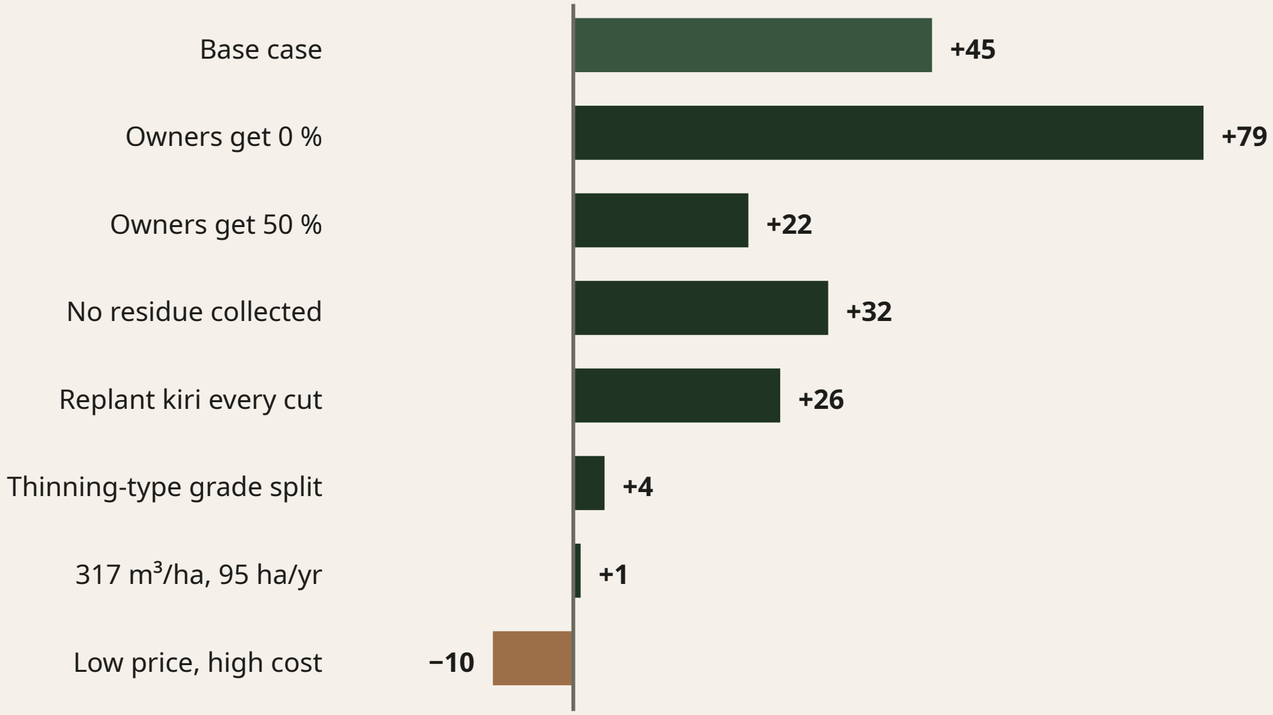
Kiri (paulownia) is a fast fuel crop on part of the land: Tokushima growth data 8.6 t dry/ha/yr; Mitsue growth data requested from SANSEI. Still checking.

THE FIRST NUMBERS

PLACEHOLDER PRICES

Our first model says plus, *then shows how fragile that is.*

STEADY-STATE SYSTEM NET, ¥M PER YEAR



Not a forecast.

50 ha/yr clear-cut at 600 m^3/ha, sawlogs sold, C+D chipped, half the land replanted to kiri, landowners paid 30% of sawlog net.

At the village's actual standing volume (277–317 m^3/ha) the base case falls to about break-even.

Left out: carbon credits, data-center income, machine purchase, wages.

Sources: kiri_sugi_integrated_system_model_2026-11-03.xlsx v1.2. All prices and costs are placeholders, not quotes.

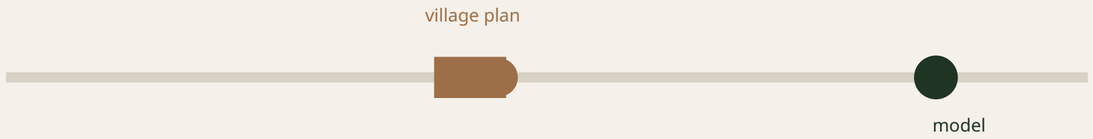
WHAT DECIDES IT

STILL RESEARCHING

Four unknowns decide the answer, *not the technology.*

Wood per hectare

M³/HA



Model assumes 600; village plan shows 277–317.

A-grade share

% OF VOLUME



Sawlog sales fund everything.

Landowner share

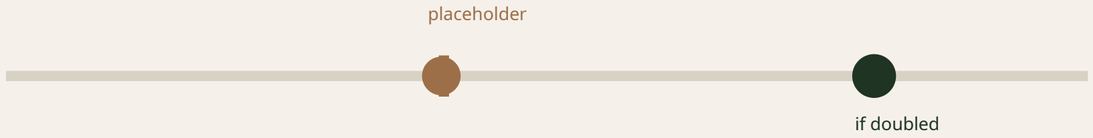
% OF SAWLOG NET



Each 10 points costs about ¥11M a year.

Residue price

¥ PER TONNE



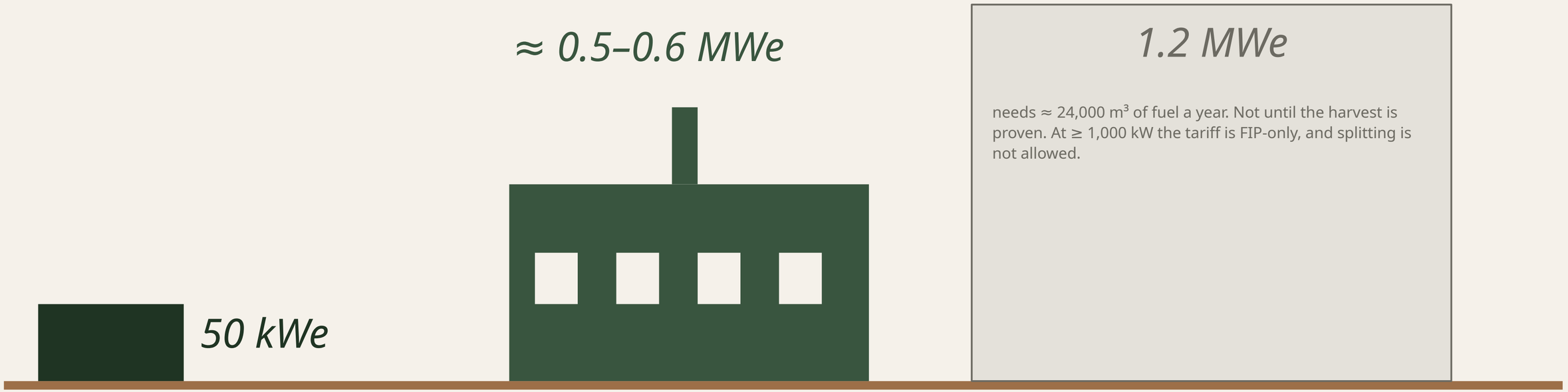
Doubling adds only about ¥1.6M a year: minor.

Sources: kiri_sugi_system_summary_2026-11-03.md, 'What drives the result'

THE ENERGY SIDE

STILL RESEARCHING

Start small to learn. *Size the real plant to the wood we can get.*



Prototype in a 20 ft container. ≈ 450–600 t dry wood a year. A learning step: efficiency only ~12–20% at this size.

First commercial size. 23–28% efficiency for gasifier plus engine at this scale. Fuel mostly C+D wood and residue.

NOT TO SCALE

Also: Japanese maker only; gasification + engine beats steam below ~1 MWe; grid at 6.6 kV has about 3 MW locally, none upstream; fuel may be bought from neighbouring villages.

Sources: Energy, finance and resources overview v1.1 (2026-11-03)



The closed Mitsue school building.
Sources: Tenkawa Forest Power Council; Aizu (2011) Safecast blog; Buena Vista, DT/Metzler, IIJ Matsue

NOT JUST US DREAMING

STILL RESEARCHING

Pieces of this already exist, *in Japan and abroad.*

Tenkawa, next door

A biomass council since 2016; a wood boiler heats the Dorogawa hot spring baths. We visited on 25 Aug 2026.

Aizu, 2011

A closed school (Akazawa Elementary) turned into a data center. A precedent for an empty school building like ours.

Wood power plus compute

Tried before: Buena Vista (US), Deutsche Telekom with Bankhaus Metzler (Germany), IIJ Matsue (Japan).

Still checking what each one really is and what transfers. Leads, not proof.

WHAT WE DID

The homework so far: *we read, we asked, we counted.*

45
village, prefectural and national acts on forestry, biomass, 協力隊 and siting, collected and summarised

113
parcels in the village thinning plan, read parcel by parcel

2
rounds of questions to Komatsu on steep-slope harvesting; gasification CHP makers compared

1.2
version of the spreadsheet model linking clear-cut, sawlogs, chips, kiri, CHP and landowner payments

5+
site and office visits: forest, village hall, Uda City cooperative, Tenkawa

100%
of it open, in a public repository, so anyone including NAIST students can check or challenge it



Forest work in Mitsue (confirm who and where).

Sources: repository docs/, research/, docs/reference/

WHAT WE DO NOT KNOW YET

STILL RESEARCHING

An honest list of open questions, *and who is answering them.*

OPEN QUESTION	WHO	STATUS
Stand volume, stumpage, who gets the sale income	**Village hall (Oba)**	asked 6 Nov
Slope limits with and without winch, road width, yearly output	**Komatsu**	follow-up sent
Real CHP efficiency and cost at 0.6–0.8 MWe	**Maker quotes**	to request
Kiri growth on Mitsue slopes	**SANSEI planting data**	asked 2 Nov
Grid connection at 6.6 kV	**Kansai T&D**	consultation drafted
Fuel bought in from neighbouring villages	**Uda City, others**	to approach
What students and researchers would measure	**NAIST (you)**	to discuss

Sources: docs/strategy/mitsue_energy_finance_resources_overview_2026-11-03.md section 5

WHERE WE STAND

Early planning. *No money raised, no contracts, no machines bought.*

NOW

Phase 0

Trust, sites, fuel talks, homework

DEC 2026

Phase 0 ends

Decision on first site and partners

2027

Prototype

50 kWe CHP and fuel trial, if site and partners settle

2028 →

Machines

A CTL trial if a fit is shown and financing is in place

LATER

Full scale

≈ 0.5–1.2 MWe and a larger compute load; no fixed date

> *¥0 committed. Dates are targets, not promises. The grant tracks we know of (METI GX regional, MoE zero-emission data center) are being studied, not applied for.*

75 years to grow the forest.

25 years to grow the village.

A hundred-year plan in the old Goryōrin spirit: planting for the people who will be here when the trees are ready.

Q&A

Tell us *where we are wrong.*

Honest critique is more useful to us than polite interest.

What would you measure? What would you doubt? Who else should we ask?



Rob Oudendijk · Mitsue-it · rob@mitsue.it

All documents open: https://codeberg.org/YR-Design/mitsue-ai-data-center

Website: mitsue.it